

UNDRALLA PURUSHOTTAMA RAO

upurush7@gmail.com | +91 8801111483 |
www.linkedin.com/in/u-purushottam | <https://github.com/hack3rslabs>

PROFESSIONAL SUMMARY

A results-driven DevOps Engineer with approximately 7 years of experience designing, implementing, and managing CI/CD pipelines, cloud infrastructure, and containerised workloads on Microsoft Azure. Proven expertise in Azure DevOps, Terraform (IaC), Docker, Kubernetes, Ansible, and automation scripting. Adept at streamlining release processes, reducing manual effort, and enabling reliable and secure software delivery at scale using Agile/Scrum methodologies.

EMPLOYMENT HISTORY

Software Engineer (DevOps)

GITAM University, Hyderabad, India

January 2022 – Present

Led DevOps implementation for large-scale digital campus systems. Key contributions include:

- Spearheaded CI/CD pipeline design and maintenance using Azure DevOps (YAML & Classic), GitHub Actions, and Jenkins across development, QA, and production environments.
- Strong proficiency in Python for automation, including experience working with APIs, structured data, and system automation libraries.
- Provisioned and managed all Azure cloud infrastructure using Terraform as Infrastructure as Code (IaC), ensuring consistent, version-controlled environments across multiple subscriptions.
- Containerised application backends using Docker (multi-stage Dockerfiles) and orchestrated deployments on Azure Kubernetes Service (AKS) with rolling update strategy and zero-downtime configuration.
- Built scalable data integration pipelines using Azure Data Factory, Synapse Analytics, and Databricks for real-time institutional reporting.
- Implemented observability stack: Azure Monitor dashboards, Log Analytics (KQL queries), Container Insights, and alert rules with Teams/email notification via Action Groups.
- Enforced security using Azure RBAC (scoped to resource group level), Azure Key Vault for secrets management, Entra ID for identity, and Microsoft Defender for Containers.
- Integrated Denodo and SQL-based data models into backend systems for operational and strategic analytics.
- Authored and maintained runbooks, deployment guides, and knowledge base articles to support operational continuity.
- Collaborated with developers, DBAs, QA engineers, and business stakeholders in Agile/Scrum sprints managed via Azure Boards and Jira.
- Managed and secured enterprise APIs using Azure API Management (APIM) with authentication and policy configuration.
- Deployed and monitored REST APIs through CI/CD pipelines, ensuring high availability and performance.

System Engineer

September 2019 – December 2021

Techwell IT Solutions, Visakhapatnam, India

- Designed and deployed data pipelines on Azure using Azure Data Factory, Synapse Analytics, and Data Lake Storage for enterprise reporting workflows.
- Automated data workflows using CI/CD pipelines built with Jenkins and Azure DevOps, reducing manual deployment effort considerably.
- Managed infrastructure provisioning with Terraform and containerisation using Docker for reliable, repeatable environments.
- Ensured code versioning with Bitbucket and collaborated with Agile teams to accelerate delivery cycles.

KEY PROJECTS

NextGen Campus Platform — Streamlined Access for Students and Staff

Client: GITAM Deemed University

Role: DevOps Engineer

Duration: April 2024 – February 2025

Skills Used: Azure DevOps, GitHub Actions, Docker, Kubernetes (AKS), Terraform, Ansible, Azure Monitor, PowerShell, YAML, Agile

A platform to centralise all digital campus services — Student Portal (attendance, fee payments, timetable, health records, hostel management), G-Learn, Moodle, and staff tools (payslips, timesheet management) into a single mobile and web application.

- Built and maintained multi-stage Azure DevOps YAML pipelines for automated builds, tests, and deployments across mobile and web platforms.
- Managed all Azure infrastructure using Terraform IaC; containerised microservices with Docker and deployed via Kubernetes with zero-downtime rolling strategy.
- Configured Azure Monitor dashboards and Log Analytics alert rules for real-time pod health, API error rate, and infrastructure capacity monitoring.
- Applied branch policies, release gates, and RBAC to enforce secure and auditable production rollouts.

PatientCare360 — Integrated Hospital Data Management System

Client: GITAM Deemed University (GIMSR Hospital)

Role: DevOps Engineer

Duration: January 2023 – March 2024

Skills Used: A fully digital patient management system for GIMSR Hospital replacing manual paper-based processes with real-time tracking of patient health records, lab reports, prescriptions, and treatment history, accessible via web and mobile.

- Designed, developed, configured, and supported enterprise APIs using Azure API Management (APIM).
- Configured APIM policies for authentication, authorization, routing, rate limiting, transformation, logging, and exception handling.
- Assisted in defining API governance standards and contributed to enterprise API architecture initiatives.
- Supported API migration projects from MuleSoft to Azure API Management.
- Developed and maintained CI/CD pipelines using Azure DevOps and GitHub Actions for automated API deployments.
- Performed code reviews, API configuration reviews, and APIM policy validations to ensure security and compliance.
- Provided technical guidance and troubleshooting support to development and operations teams.
- Configured API monitoring, logging, and alerting using Azure Monitor, Application Insights, Log Analytics, and ELK Stack.
- Diagnosed and resolved API, networking, security, performance, and production-related issues.
- Supported SIT, UAT, production deployments, release validation, and post-deployment activities.
- Prepared technical design documents, deployment guides, operational runbooks, and knowledge base articles.
- Managed Incident, Problem, Change, and Release Management activities following ITIL best practices.
- Collaborated with solution architects, Scrum Masters, DevOps, Security, Infrastructure, and Application teams to deliver enterprise API solutions.
- Identified process improvements and implemented automation to enhance API performance, reliability, and operational efficiency.

Campus AD+ — Active Directory Automation for Multi-Campus Infrastructure

Client: GITAM Deemed University

Role: DevOps Engineer

Duration: March 2022 – December 2022

Skills Used: Azure DevOps, Terraform, PowerShell, Azure AD / Entra ID, Azure Monitor, Log Analytics, RBAC, YAML, Git, Windows Server 2012 R2

Automated and standardised Active Directory integration across three GITAM campuses for over 10,000 systems, reducing manual provisioning effort by 80%.

- Authored bulk PowerShell automation scripts for Active Directory domain joins, OU placement, and machine provisioning; processed machines in parallel batches.
- Designed Terraform modules for domain controller VMs, DNS, and AD infrastructure across three campuses using IaC principles.
- Built Azure DevOps pipelines for AD component deployment with campus-level manual approval gates between rollout phases.
- Integrated Azure Monitor and Log Analytics for real-time directory health, login activity, and sync failure alerting.
- Implemented RBAC policies and Azure Policy to restrict administrative access and satisfy internal audit and compliance standards

TECHNICAL SKILLS

AI Tools: GitHub Copilot, ChatGPT / OpenAI, Claude (Anthropic), Azure OpenAI Service — used for code generation & review, pipeline/IaC script assistance, and documentation writing

Cloud Platforms: Azure (AKS, Data Factory, Key Vault, Monitor, Entra ID, Defender for Cloud, ACR, VNets, NSGs, Synapse Analytics, Backup), AWS

CI/CD & Automation: Azure DevOps, GitHub Actions, Jenkins, GitLab CI/CD, Harness, Bamboo

IaC & Config Management: Terraform, Ansible, Salt, ARM Templates

Containerisation: Docker, Kubernetes/AKS, Helm, Azure Container Registry

Security & Identity: Azure RBAC, Key Vault, Entra ID, PIM, Defender for Cloud, NSG, Private Endpoints

Monitoring: Azure Monitor, Log Analytics (KQL), Container Insights, Splunk, AppDynamics

Scripting: PowerShell, Bash/Shell, Python, YAML

Version Control: Git, GitHub, Azure Repos, Bitbucket

Databases & Data: Azure Data Factory, Synapse Analytics, Databricks, Denodo, Oracle, MySQL

Build Tools: Maven, Azure Artifacts

Operating Systems: Linux (RHEL, Ubuntu, CentOS), Windows Server

Methodologies: Agile, Scrum, Jira, Azure Boards, Zendesk

EDUCATION

Master of Science in Information Technology (MSc IT)

Year of Completion: 2020

Department/Field: Information Technology

CERTIFICATIONS

Microsoft Azure Certifications

Azure Administrator Associate — AZ-104

Issuing Authority: Microsoft

Status: Certified

Description: Validates technical proficiency in managing, implementing, and monitoring an organisation's Microsoft Azure environment, including virtual networks, storage, compute resources, identity, and governance.

Azure Security Engineer Associate — AZ-500

Issuing Authority: Microsoft

Status: Certified

Description: Covers implementation of security controls, maintaining an organisation's security posture, identifying and remediating security vulnerabilities, and implementing threat protection across Azure services.

Microsoft Certified Solutions Associate (MCSA)

MCSA 70-410 — Installing and Configuring Windows Server 2012

Issuing Authority: Microsoft

Status: Certified

Description: Covers installation, configuration, and administration of Windows Server 2012, including Active Directory Domain Services, DNS, DHCP, Group Policy, and Hyper-V.